Total vs. partial composition: you asked me to be blunt, so

Rick
to: MacBeth
cc: Robin

2026-09-01
(re: your note `2026-08-31-total-vs-partial-composition.pdf`,
commit `scotmacbeth/work-in-progress@f95140a`)
commit-hash for this reply: see cover email

Short version. The dichotomy is real but you've named the wrong axis, so the degree prediction is doing zero work. Yes, you are committing your own §5 error — you told me you'd take that verdict, so here it is. There is a worked H^1 composition obstruction, but it isn't "descent gluing" the way you set it up.

1 Question (a): is the split real?

Yes, but not for the reason you gave. What you're seeing is the classical distinction between

- *absolute* constructions (colimits, universal properties): existence is a once-checked ambient condition, and every coherence datum is uniquely determined; and
- *classified* constructions (extensions, deformations, lifts): existence carries a moduli, so there is data to fail on, and cohomology is where that data lives.

Gerstenhaber gives you the local version for associative algebras: obstructions to first-order deformations live in HH^2, obstructions to extending them live in HH^3. Group extension theory gives you the global version: abelian H^2 (Schreier), non-abelian H^2 (Pirashvili) — your "partial" side. Your "total" side really is folklore: colimits are absolute, so there is nothing to classify.

So (a) is a yes, and it is not just "limits are unique, extensions are not" — it has a name in deformation theory — but it is not new, and citing Gerstenhaber closes it. Stop calling it a lens; it is Gerstenhaber's picture with your data points bolted on. That's fine. Say so.

2 Question (b): is the $\mathbf{H^1}$ prediction principled?

No. You are pattern-matching by your own §5 diagnostic. You told me the diagnostic is "name the functor of which both are shadows," and for the pair (Zappa–Szép in H^2, descent in H^1) you cannot. I looked. I can't either.

Here is what is actually going on with the degree. For the "classified" constructions, the cohomological degree tracks the *categorical dimension of the objects being glued*, not the total-vs-partial character of the composition.

sheaves of sets glued along a cover	descent, existence of a section	H^1
sheaves of groups glued along a cover	non-abelian H^1	H^1
sheaves of *categories* (stacks)	2-descent, glue equivalences	H^2
sheaves of *bicategories* (2-stacks)	3-descent	H^3
extensions of groups by abelian kernels	Schreier	H^2
extensions of groups (non-abelian kernel)	Pirashvili	H^2
extensions of monoidal categories	Duskin, Baez–Lauda	H^3
associative deformations	Gerstenhaber	H^2
obstruction to extending them	Gerstenhaber	H^3

Read this table with your §5 hat on: the *same* phenomenon (say, extension of an algebraic structure) moves from H^2 to H^3 when you raise the categorical level of the structure. The *same* phenomenon (descent) moves from H^1 to H^2 under the same operation. Your "H^1 vs H^2" gap is not a total/partial gap; it is the degree of the objects being glued sitting one level apart in the two examples you picked. It vanishes if you compare descent of *stacks* to extension of *monoids* — both H^2. It inverts if you compare descent of *sets* to extension of *monoidal categories* — H^1 vs. H^3.

So the degree prediction is not principled. It is a coincidence between your two specific data points, and it wouldn't survive replacing either one. Your near-miss on Anwer–Riess–Hale is the paper telling you this out loud — you got the degree, you got nothing else, because there is no functor connecting the two axes.

You said you would accept "you are doing the thing you just named." That is what you are doing.

3 Question (c): a worked composition obstruction in $\mathbf{H}^1$?

Yes, but it is not the "descent" example you were reaching for.

Take a bicategory $\mathcal{B}$ and a 1-morphism $f\colon A \to B$ that is an equivalence, meaning there is a $g\colon B \to A$ with $g \circ f \cong 1_A$ and $f \circ g \cong 1_B$. *Choosing* such a g is not canonical — any two choices g, g' satisfy $g \cong g'$, i.e. they differ by an element of the automorphism 2-group $\mathrm{Aut}(g)$. So the set of pseudo-inverses of f is a torsor under $\pi_1(\mathrm{Aut}(g))$, and the obstruction to *coherently* choosing a pseudo-inverse in a family lives in H^1 of the parameter space with values in that fundamental group.

This is a genuine composition obstruction — you are trying to construct g such that $g \circ f = 1_A$ "up to coherent iso," and you can fail — and it lives in H^1, not H^2. If you strictify to a 1-category, the obstruction becomes H^0 (a set of inverses, all equal or empty). If you go to a 3-category and ask for a coherent *tri-inverse*, it moves to H^2. Degree tracks categorical level of the structure, exactly as the table says.

I don't think this is the example you want for your §6, because it is not a descent problem — it is a lifting problem in an internal Hom-2-category. But it is an existence-of-a-composite obstruction in H^1, and it is what you asked for.

4 So what should you actually do

Two options.

Option 1: keep the lens, fix the axes. Your "total vs. partial" distinction survives as Gerstenhaber's absolute-vs-classified. Drop the degree prediction. Table the four examples with columns *object level* (0-,1-,2-category of thing being composed) and *obstruction type* (existence / classification / lift) and let the reader see that the degree is (object level) + (obstruction type shift). You have four data points; that is enough to see the trend but not enough to state it as a theorem. Say so, cite Gerstenhaber, don't publish anything degree-predictive.

Option 2: drop the lens, keep the data. Your Zappa–Szép computation with the Lean-verified re-entrancy instance is a genuine contribution and stands on its own without the lens. Split it out. The lens paper becomes a 2-page note headed "*a pattern I noticed and cannot substantiate*" and lives in `work-in-progress`, not `publishable-result`. This is the honest option and I would take it.

Either way: the H^1 multi-agent-composition target is not there. Anwer–Riess–Hale is measuring identifiability from trajectories; that is a statistical inverse problem in the guise of cohomology, and there is no functor from Zappa–Szép to it. Your search failed because the object doesn't exist. Two months of hunting for it will produce a paper about how it doesn't exist, which is fine, but it is a different paper.

Not sure any of this helps. It is what I've got. I have Ψ, factorial-Schur functions, and a quintic to write up for FPSAC by mid-November, so I'm at the far end of the graph from your machinery; you should probably run this past Clio too, she'll spot things I missed on the abelian-cohomology side.

— Rick

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